
MODEL SELECTION AND VALIDATION REPORT (remove_1_biomass model)

OBSERVATIONS:1732

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_1_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.89	0.42
remove_1_kde_fric	-0.10	0.03
remove_1_kde_feve	-0.10	0.03
sbt_mean	0.27	0.09
sbt_sd	0.05	0.04
sst_sd	0.04	0.04
log_chla_mean	0.04	0.03
depth	-0.03	0.04
log_sum_fishing	0.10	0.03

Dispersion parameter: 0.83

Spatiotemporal AR1 correlation (rho): 0.94

Matérn range: 501.99

Spatiotemporal marginal AR1 SD: 0.89

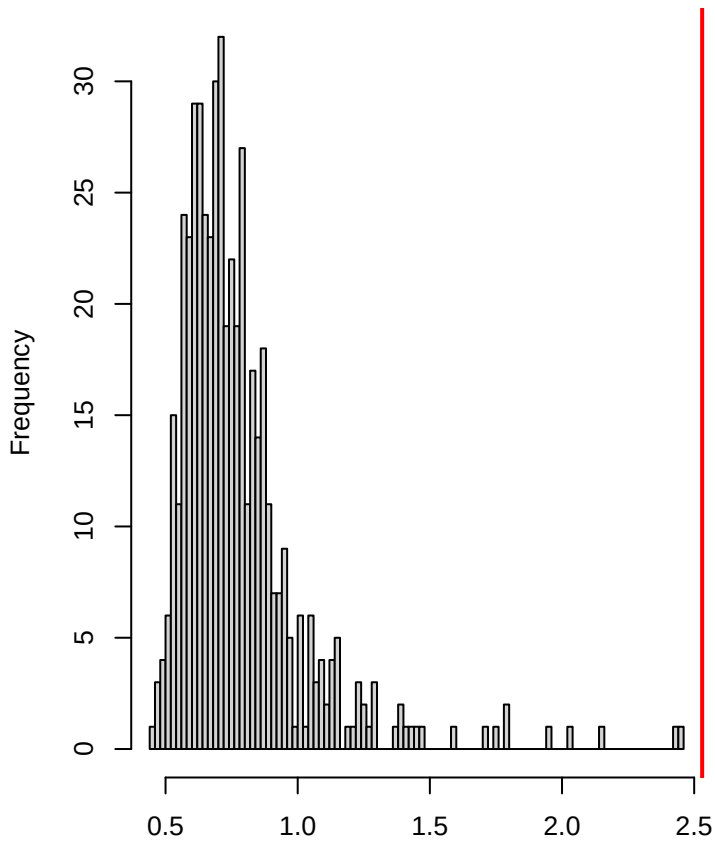
REML criterion at convergence: 13345.547

See ?tidy.sdmTMB to extract these values as a data frame.

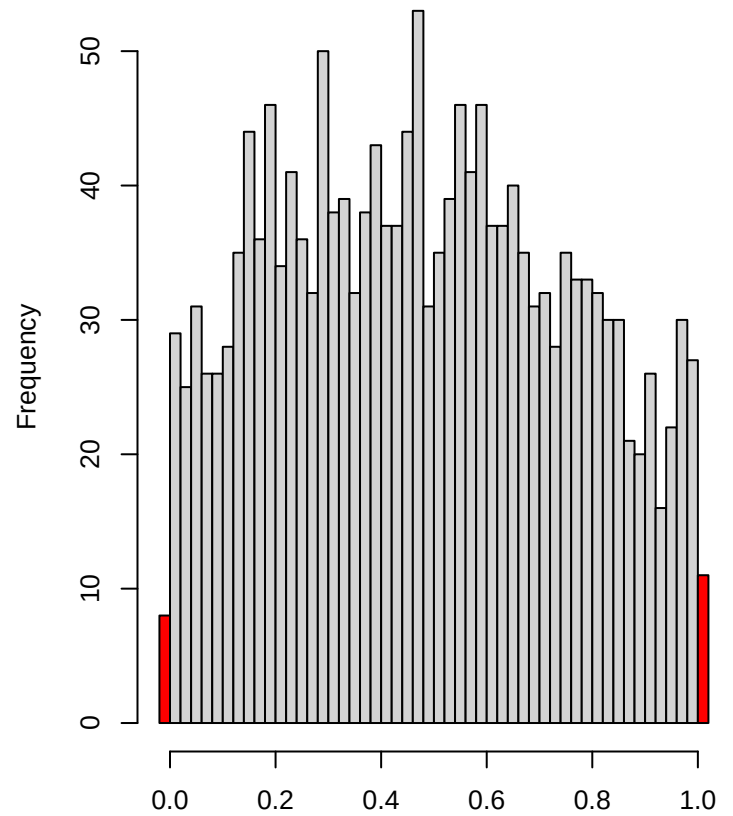
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	26717.09	3.880500
spatiotemporal_random_fields_ar1_with_anisotropy	15	26720.97	117.927177
spatiotemporal_random_fields_iid	12	26838.90	1.504629
spatiotemporal_random_fields_iid_with_anisotropy	14	26840.41	33.721567
spatial_random_fields	12	26874.13	3.716616
spatial_random_fields_with_anisotropy	14	26877.84	239.742623
error_only	10	27117.59	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



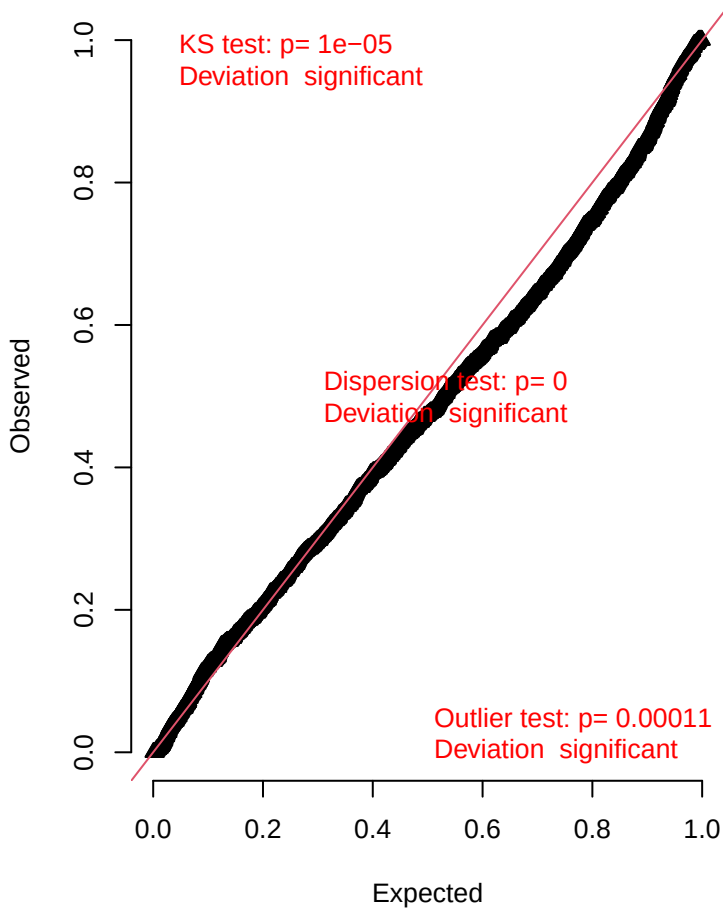
Outlier test significant



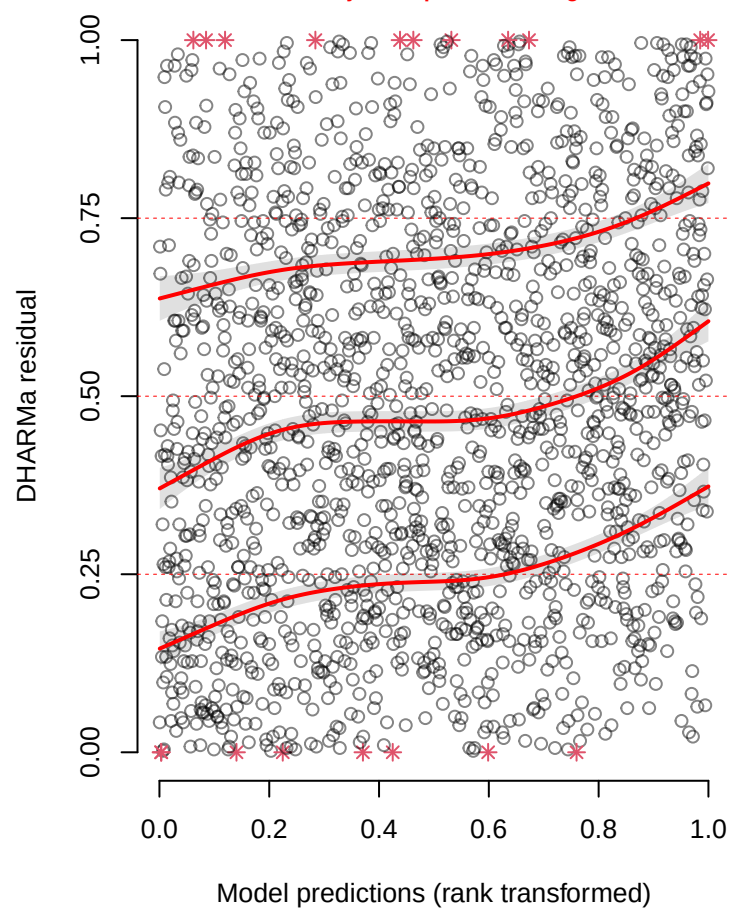
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.039 | H1: Distance-based autocorrelation | p-value = 0

